

Boundary-Aware Agricultural Field Segmentation

Comparing direct weighting, auxiliary distance, and explicit boundary prediction

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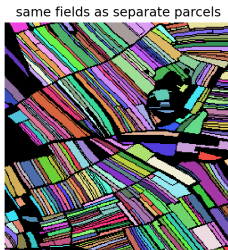
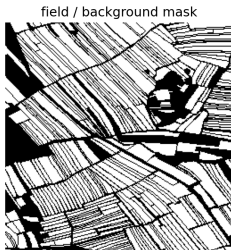
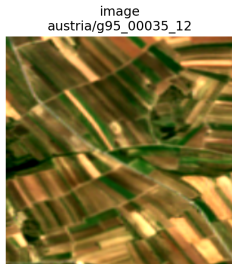
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Four controlled methods

1,430 held-out test chips

The challenge is preserving individual fields, not only field area



Failure mode

A binary mask can cover the correct area while merging neighboring fields.

Project goal

Improve border agreement without sacrificing region overlap.

Source: FTW imagery and labels; visualization generated in this project.

Data and supervision

Fields of The World subset

Countries	Austria, Croatia, Denmark
Input	two seasonal images, each with RGB+NIR (8 channels total)
Resolution	256 × 256 pixels; 10 m per pixel
Labels	field mask + FTW instance mask

10,938 train

1,347 validation

1,430 test

How the labels are used

Binary field mask

Mask head

Direct target: field or background at every pixel.

FTW instance mask

Provided by the dataset; separates neighboring fields.

converted into ↓

Signed distance field (SDF)

Distance head

Derived target: distance to the nearest field boundary.

Source: Kerner et al., "Fields of The World," AAAI 2025.

Related work and what we contributed

Built upon

- ▶ **FTW**: global, multi-date Sentinel-2 field benchmark.
- ▶ **U-Net**: encoder–decoder with skip connections for dense prediction.
- ▶ **Distance-aware supervision**: geometry can complement region losses.

Our project work

- ▶ **Boundary-weighted BCE**: emphasized mask errors near field borders.
- ▶ **SDF regression**: added a distance head with boundary-focused supervision.
- ▶ **Boundary prediction**: added a second head that classifies border pixels.

Controlled comparison: same U-Net and held-out test set.

Research question: where should boundary information enter the learning objective?

Source: Ronneberger et al., MICCAI 2015; Kervadec et al., MIDL 2019 / MedIA 2021.

Four methods, one controlled comparison

Method	Architecture	Training loss	Boundary signal
Mask baseline	one mask head	$\mathcal{L}_{BCE} + \mathcal{L}_{Dice}$	none
Boundary BCE	one mask head	$\mathcal{L}_{WBCE} + \mathcal{L}_{Dice}$	direct mask gradients
Two-head SDF ($\lambda = 0.5$)	mask + distance heads	$\mathcal{L}_{BCE} + \mathcal{L}_{Dice} + 0.5\mathcal{L}_{WSDF}$	weighted auxiliary geometry
Mask + boundary	two active heads	$\mathcal{L}_{WBCE} + \mathcal{L}_{Dice} + 2\mathcal{L}_{Bnd}$	mask gradients + explicit border task

Held constant

Data, augmentation, U-Net width, optimizer, and evaluation code.

Checkpoint audit

The trained mask baseline uses ordinary **BCE + Dice**.

Method 1: put boundary emphasis directly into BCE

Boundary-weighted BCE

$$\mathcal{L}_{WBCE} = \frac{\sum_i w_i \text{BCE}(z_i, y_i)}{\sum_i w_i}$$
$$w_i = 1 + 20 \exp\left(-\frac{|D_i|}{0.12}\right)$$

$$\mathcal{L}_{final} = \mathcal{L}_{WBCE} + \mathcal{L}_{Dice}$$

Dice preserves global region overlap; weighted BCE changes which pixel errors dominate the gradient.

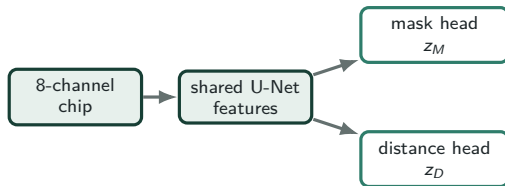
Pixel weight by distance

$ D $	BCE weight
0	21.0
0.12	8.36
0.24	3.71
far from border	≈ 1.0

Key difference

The boundary signal acts on the **final mask output**, not through an auxiliary task.

Method 2: boundary-focused signed-distance regression



Hypothesis

The SDF task forces shared features to encode individual field geometry.

- ▶ $D > 0$: inside an individual field
- ▶ $D \approx 0$: field boundary
- ▶ $D < 0$: background
- ▶ Inference still uses the mask head

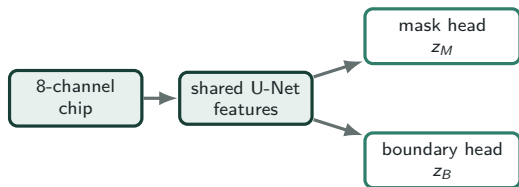
+9,313 parameters

+0.12%

Joint objective

$$\mathcal{L}_{2H} = \mathcal{L}_{BCE} + \mathcal{L}_{Dice} + 0.5 \mathcal{L}_{WSDF},$$
$$\mathcal{L}_{WSDF} = \frac{\sum_i a_i \text{SmoothL1}(\tanh(z_{D,i}), D_i)}{\sum_i a_i},$$
$$a_i = e^{-|D_i|/0.12}$$

Method 3: a second head predicts boundary pixels



Why add this head?

Boundary-weighted BCE focuses the mask errors; the second head explicitly teaches the shared features which pixels are borders.

Two-head training objective

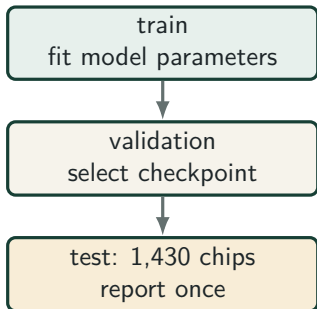
$$B_i = 1[|D_i| \leq 0.12]$$

$$\mathcal{L}_{M+B} = \mathcal{L}_{WBCE}(z_M, Y) + \mathcal{L}_{Dice}(\sigma(z_M), Y) + 2 \mathcal{L}_{BCE}^{bal}(z_B, B)$$

- ▶ $B = 1$: close to a field boundary
- ▶ $B = 0$: all other pixels
- ▶ Class balancing handles the thin positive boundary band
- ▶ Segmentation still comes from the mask head

Two active output heads

Evaluation protocol: model selection and test stay separate



Fair comparison

- ▶ Common probability threshold: **0.60**
- ▶ Boundary band:
2 px inward from each mask edge
(erosion radius 2; ≈ 20 m)
- ▶ Same 1,430 test chips for all models

Metrics

mIoU: region overlap

Boundary IoU: overlap of two-pixel inner bands extracted from predicted and target masks

The explicit boundary head gives the best two-pixel Boundary IoU

Mask baseline		0.3469
Boundary BCE		0.3926
SDF head ($\lambda = 0.5$)		0.3501
Mask + boundary		0.4000

Inner-boundary IoU (2 px) @ threshold 0.60

Method	mIoU	BloU
Baseline	.6572	.3469
Boundary BCE	.6675	.3926
SDF head ($\lambda = 0.5$)	.6499	.3501
Mask + boundary	.6634	.4000

Main test finding

The explicit boundary head adds **+0.74 pp BloU** over Boundary BCE alone; Boundary BCE alone keeps the highest mIoU.

Direct boundary supervision gives the larger gain

All four test checkpoints

Test @ 0.60	mIoU	BloU
Baseline	.6572	.3469
Boundary BCE	.6675	.3926
SDF head ($\lambda = 0.5$)	.6499	.3501
Mask + boundary	.6634	.4000

What changed

Boundary-weighted mask loss

Directly changes mask gradients; +4.57 pp BloU.

Distance regression

Boundary-focused auxiliary task; +0.32 pp BloU, but -0.73 pp mIoU.

Explicit boundary head

Adds a focused border task; another +0.74 pp BloU.

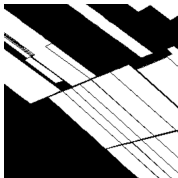
SDF gives a small boundary gain; direct mask and boundary supervision gives much more.

Qualitative comparison on the same Austria chip

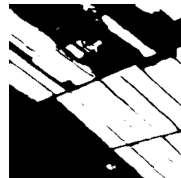
Sentinel-2 RGB



Ground truth



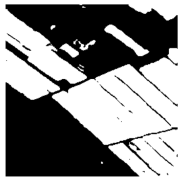
Baseline



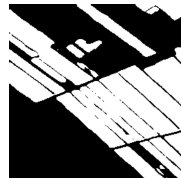
Boundary BCE



SDF auxiliary head ($\lambda = 0.5$)



Mask + boundary heads



Same image and probability threshold (0.60) for every prediction.

Source: FTW validation chip [austria/g83_00040_14](#); predictions generated in this project.

Conclusion and outlook

Takeaway

Direct boundary supervision is more effective than auxiliary signed-distance regression.

- ▶ Boundary BCE: 0.3469 $\rightarrow$ 0.3926 BIoU (**+4.57 pp**)
- ▶ Adding the boundary head reaches **0.4000 BIoU (+5.31 pp)**
- ▶ New model mIoU is 0.6634, still +0.62 pp above baseline
- ▶ SDF with $\lambda = 0.5$: **+0.32 pp BIoU**, but -0.73 pp mIoU

Best boundary model: boundary-weighted mask loss + explicit boundary head.

Next steps

- ▶ tune loss weights and hyperparameters
- ▶ report per-country performance
- ▶ test instance-separation post-processing
- ▶ train and evaluate on more data

Backup: threshold sensitivity on the test set

Method	0.40		0.50		0.60		0.70	
	mIoU	BloU	mIoU	BloU	mIoU	BloU	mIoU	BloU
Baseline	.6593	.3196	.6587	.3354	.6572	.3469	.6506	.3506
Boundary BCE	.6706	.3533	.6731	.3790	.6675	.3926	.6520	.3850
2-head SDF ($\lambda = 0.5$)	.6586	.3282	.6549	.3414	.6499	.3501	.6393	.3497
Mask + boundary	.6717	.3665	.6685	.3875	.6634	.4000	.6467	.3875

Why the main talk uses 0.60

It is a single common operating point for all four models. The new model has the best Boundary IoU at every tested threshold.

Backup: exact objectives

Boundary BCE

$$\mathcal{L}_{BBCE} = \mathcal{L}_{WBCE}(z_M, Y) + \mathcal{L}_{Dice}$$

One mask output; $\alpha = 20$ and $\sigma = 0.12$ define the pixel weights.

Mask + boundary heads

$$\mathcal{L}_{M+B} = \mathcal{L}_{WBCE} + \mathcal{L}_{Dice} + 2\mathcal{L}_{BCE}^{bal}(z_B, B)$$

Two active outputs; the explicit boundary term has weight 2.

How the boundary target is formed

$B_i = \mathbf{1}[|D_i| \leq 0.12]$. The precomputed distance target constructs the weights and boundary labels; the final model predicts only the mask and boundary maps.

References

-  H. Kerner et al.
Fields of The World: A Machine Learning Benchmark Dataset for Global Agricultural Field Boundary Segmentation.
AAAI, 2025.
-  O. Ronneberger, P. Fischer, and T. Brox.
U-Net: Convolutional Networks for Biomedical Image Segmentation.
MICCAI, 2015.
-  H. Kervadec et al.
Boundary Loss for Highly Unbalanced Segmentation.
MIDL, 2019; extended in Medical Image Analysis, 2021.

All prediction panels and project diagrams were generated from this repository; no external figures are reproduced.